Next Steps Brainstorm: Beyond the Honest Null

Where We Stand

The evidence is now clear and well-diagnosed:

Finding	Evidence
Teacher privilege gap exists	Gemma-4: +0.07 F1 (with_solution 0.786 vs no_gt 0.716)
Labels genuinely diverge	Only 69.1% step-level agreement; 65% of solutions have ≥1 disagreement
Student fails to absorb privilege	Paired BoN: nogt (37.5%) ≥ priv (34.0%), p=0.14; neither beats MV (39.0%)
Neither student beats majority vote	prm_rerank < majority_vote across all setups

The bottleneck is diagnosed: the teacher *has* the signal, the labels *carry* the signal, but the **1.5B student can't utilize it**. This maps to the roadmap's "capacity bottleneck" regime.

> [!IMPORTANT] The gating weakness from the roadmap remains: **until the student beats majority vote, the transfer null is confounded with "we trained a weak PRM."** This is P0.

Approach 1: Student Capacity Sweep (Phase C1)

Hypothesis: Privilege transfer is capacity-gated — it will emerge when the student is large enough to represent the teacher's nuanced error-detection features.

What to do:

- Train identical priv vs nogt students at **0.5B, 1.5B (done), 3B, and 7B** parameter counts
- Use the same Gemma-4 teacher labels (already generated for N=1000)
- Candidates: `Qwen2.5-0.5B-Instruct`, `Qwen2.5-3B-Instruct`, `Qwen2.5-7B-Instruct`

Expected outcome:

- If privilege gap **opens** at 3B or 7B → strongest paper framing: "when does privileged supervision distill?" with a clean capacity boundary
- If gap stays flat at 7B → privilege genuinely doesn't transfer via score distillation (mechanism problem, not capacity)

Effort: Medium. Reuse existing labeled data. The 7B student needs careful VRAM management (LoRA + gradient checkpointing on 2×3090), but 3B should be straightforward.

Risk: The 7B student may OOM during training. The sweep takes ~4× the current training time but training is fast relative to generation/labeling.

Approach 2: Scale Training Data (Phase B1)

Hypothesis: N=1000 is insufficient for the student to learn the subtle privileged signal. The privilege information is a "fine detail" that requires more data to separate from noise.

What to do:

- Scale N_TRAIN from 1k → **5k** → **10k**
- This requires regenerating solutions (local gemma-2-9b-it) and relabeling (remote Gemma-4 teacher)
- Also sweep learning rate and epochs (LR ∈ {1e-4, 5e-5, 1e-5}, epochs ∈ {2, 4, 8})

Expected outcome:

- More data should first improve the *absolute* student quality (hopefully beating MV — solving P0)
- Then reveals whether the priv-nogt gap opens with more data

Effort: High. Generation at N=5k requires ~5× the current generation time (~7.5 hours). Labeling through Edward's MacBook at ~32k requests for N=1k means ~160k requests for N=5k — potentially 2+ days of remote labeling.

Risk: Generation/labeling pipeline is the bottleneck, not training. May need to coordinate long labeling windows with Edward. If the student is still below MV at N=10k, the bottleneck is architectural, not data.

Approach 3: Positive Control Experiment (Phase C3)

Hypothesis: The distillation pipeline itself is sensitive to teacher quality — privilege is specifically what fails to transfer, not "everything."

What to do:

- Train students from three teachers:
    1. **Strong teacher** (Gemma-4 + GT solution) — existing priv labels
    2. **Weak teacher** (Gemma-2-9b, no GT) — a deliberately weaker labeler
    3. **Label-shuffled** (randomly shuffle the priv labels) — calibration baseline
- If student quality tracks: strong > weak > shuffled → pipeline works, privilege null is real
- If strong ≈ weak ≈ shuffled → pipeline is insensitive (back to B)

Expected outcome: This single experiment is what the roadmap calls "what convinces a skeptical reviewer the null means something." If the pipeline IS sensitive to teacher quality but NOT to privilege specifically, the null result becomes far more credible and publishable.

Effort: Low-Medium. Relabeling with gemma-2-9b-it (local, fast, no remote dependency). Shuffled labels are trivial. Training is cheap. Can run in 1 day.

Risk: Very low — this is a diagnostic, not a new technique. But it doesn't make the student *better*, it only makes the null *more defensible*.

Approach 4: Change the Distillation Method (New Technique)

Hypothesis: The current distillation approach (MSE on a scalar score head) is too lossy to capture the privileged signal. The privilege lives in the *structure* of the teacher's reasoning, not just the score.

Sub-approaches:

4A: Distribution Matching Instead of Point Estimates

- Instead of MSE on the teacher's scalar score, use **KL divergence against the teacher's full token-level logit distribution** over the critique
- The privileged teacher's *reasoning chain* carries the signal, but `scoring_loss` (MSE on a single float) throws it all away
- Currently the `logit_loss` is **disabled** because vocabs differ (Qwen student vs Gemma teacher). Fix this with a **vocabulary projection layer** or use a **same-family** student (Gemma-2-2B)

4B: Contrastive Distillation

- Instead of training on labels independently, train on **pairs**: for the same step, present the priv label AND the nogt label, and train the student to predict which came from the privileged teacher
- This forces the student to model the *difference* that privilege makes, rather than fitting a noisy target
- Implementation: triplet/contrastive loss where anchor = step, positive = priv score, negative = nogt score

4C: Step-Level Critique Distillation with Chain-of-Thought

- The current `priv_critique` condition trains the student to *generate* the teacher's critique text (via `lm_loss`). But the critique text is generated by a 26B model — the 1.5B student may not have the capacity to replicate it
- Instead: **extract the binary verdict** from the teacher's critique (correct/incorrect + reason category) and train on structured labels rather than free-text
- Alternatively: use the teacher's critique as a **chain-of-thought prompt** — feed it to the student at training time and train only the score head conditioned on seeing the critique

Effort: Medium-High. 4A requires architectural changes. 4B is novel and needs new training code. 4C is the most practical and could be implemented as a data preprocessing change.

Risk: These are more speculative. 4B in particular is novel but unproven. The risk of over-engineering is real — the current pipeline already works for strong-vs-weak teacher signals (pending C3 verification).

~~Approach 5: Symbolic MATH Checker + Better BoN Eval (Phase B2)~~ [ALREADY IMPLEMENTED]

Correction: This approach was already implemented. The `answers_match` function in `scripts/generate_solutions.py` uses `math_verify` and is correctly imported by both `bon_rerank.py` and `bon_paired.py`. The "null" and "below MV" results in Phase 3 are accurate and already use symbolic checking.

Approach 6: Reframe as Teacher-Only Paper (Fallback)

Hypothesis: N/A — this is a framing pivot, not an experiment.

What to do:

- Accept that distillation into a small student is not the contribution
- Publish the **teacher-level sweet spot** as the main result:
  - 3-tier difficulty curve (GSM8K ≈0, MATH +0.05, OlympiadBench ≈0)
  - Rescue mechanism (self-verification failure × tractability)
  - Cross-family validation (Qwen-27B +0.082)
  - The evidence pack with by-MATH-level breakdown (L1 -0.13 → L3 +0.11 → L5 +0.04)
- The "null transfer" becomes a secondary finding: "better labels don't automatically mean better students"

Effort: Low (mostly writing, no new compute).

Risk: Lower impact/novelty — the teacher sweet-spot alone may be a workshop paper rather than a main venue paper. But it is honest, citable, and ready now.

Comparison Matrix

Approach	Solves P0?	Tests Transfer?	Effort	Risk	Paper Impact	Prerequisite
1. Capacity sweep	Partially (7B might beat MV)	☑ Directly	Medium	Medium (OOM)	🔥🔥🔥 Highest if gap opens	None
2. Scale data	Likely (more data → better PRM)	☑ Indirectly	High	Medium (labeling time)	🔥🔥	Edward's endpoint
3. Positive control	❌ No	☑ Validates null	Low-Med	Low	🔥🔥 Defensibility	None
4. New distillation	Maybe	☑ New technique	Med-High	High (speculative)	🔥🔥🔥 If works	Depends on sub-approach
~~**5. Symbolic checker**~~	N/A	N/A	N/A	N/A	Done	None
6. Reframe	N/A	N/A	Low	Low	🔥 Workshop-level	None

Recommended Priority Ordering

Tier 1 — Do Now (cheap, unblocks defensibility)

> [!TIP] This takes ≤1 day.

1. **Approach 3 (Positive control)** — cheapest experiment that makes the null publishable. Train a student from a deliberately weak teacher (e.g. `gemma-2-9b` locally) vs your existing Gemma-4 labels. If the student tracks teacher quality but not privilege specifically, the null becomes defensible and publishable. Regardless of what else you do, you need this for the paper.

Tier 2 — The Big Swing (pick one)

2. **Approach 1 (Capacity sweep)** — this is the roadmap's top recommendation and the most likely path to the **strongest paper** ("when does privilege distill?"). If a 7B student shows the gap, you have a clean capacity boundary. Start with 3B (fits easily on 2×3090 with LoRA) as a quick signal before committing to the full sweep.

Tier 3 — If Capacity Sweep Shows No Gap

4. **Approach 4C (Structured critique distillation)** — if the gap doesn't open at 7B, the mechanism is the bottleneck, not capacity. Try structured labels before going fully novel.

5. **Approach 2 (Scale data)** — only if 4C doesn't work and you have the labeling bandwidth.

Tier 4 — Fallback

6. **Approach 6 (Reframe)** — only if everything above fails. You already have a publishable teacher-only result.

Open Questions for You

1. **Is Edward's Gemma-4 endpoint still available?** Approaches 1 and 2 reuse existing labels, but if you want fresh labels at larger N, you need the endpoint.

2. **VRAM budget for 7B student training:** Have you tried loading `Qwen2.5-7B-Instruct` with LoRA on the 3090s? With 4-bit base + LoRA adapters + gradient checkpointing, it should fit in ~20GB, leaving headroom. Worth a quick test.

3. **Does the team prefer "proving the null is real" (Approach 3) or "finding where it breaks" (Approach 1)?** Both are valid paper framings but lead to very different stories.